

Tarit Witworrasakul

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Available for Co-op: Summer 2026 – Fall 2026

EDUCATION

University of British Columbia

September 2023 – May 2028

Bachelor of Science in Computer Science with minor in Commerce

Vancouver, BC

- Outstanding International Student Award

WORK EXPERIENCE

CPSC 310 Undergraduate Teaching Assistant

September 2025 – April 2026

University of British Columbia

Vancouver, BC

- Worked as part of a teaching team supporting 749 students in the Introduction to Software Engineering course

Software Engineer Co-op

May 2025 – August 2025

Agoda Services (subsidiary of Booking Holdings Inc.)

Bangkok, Thailand

- Developed a load testing suite using Grafana k6 with randomized input data, simulating up to 500 req/sec and logging metrics to Grafana dashboards to proactively validate system performance before new feature rollouts
- Stabilized GitLab CI/CD pipeline by integrating MockAPI endpoints to fix flaky integration tests and upgrading dependencies to address security vulnerabilities, ensuring tests run on every merge request
- Improved Cashback processing logic in a C# (.NET Core) API by creating SQL stored procedures and integrating a GraphQL API, automating cases that previously required manual review
- Resolved production issues by adding retry logic with fallback flags to handle API spec edge cases, improving reliability of the Cashback page

Software Developer Intern

June 2024 – August 2024

Applicad Public Company Limited

Bangkok, Thailand

- Developed a real-time facial recognition web application using Next.js for customer identification via webcam
- Built a backend service in Python with FastAPI for real-time facial recognition, using ChromaDB for face vector comparison and PostgreSQL with SQLAlchemy ORM for user data, integrated with the existing internal database
- Deployed the application and databases to an on-premises Ubuntu server using Docker and Docker Compose

EXTRACURRICULARS

MLH Meta Production Engineering Hackathon - Reliability Track Winner 🏆 | *Python, K8s* April 2026

- Built a URL shortener API using Flask and Valkey caching, reducing response times for repeated lookups
- Provisioned a 3-node K8s cluster on bare-metal Dell Optiplex, deploying the app as 3 replicas behind Traefik ingress with high-availability PostgreSQL (CloudNativePG, streaming replication) and Valkey sentinel caching
- Designed CI/CD pipelines in GitHub Actions with service containers and diff-coverage checks on PRs, automated container builds to GHCR, and deployment to K3s via a self-hosted runner on merge to main
- Built an observability stack with Prometheus, Grafana dashboards, Loki log aggregation, and Alertmanager

BitRelay - EDT Design Competition 2026 - Best Social Impact Award Winner 🏆 | *Python* March 2026

- Built an ESP32 BLE relay node for bitchat using MicroPython and aioBLE, extending Bluetooth mesh range for communication during internet shutdowns and disasters

Ring Notify 🏆 | *Golang, TypeScript, React Native, PostgreSQL, Azure*

October 2023 - Feb 2026

- Built REST APIs for user registration and IoT-triggered notifications using Golang (Echo) and Firebase Cloud Messaging; automated deployment to Azure VM via Docker and GitHub Actions

UBC Uncrewed Aircraft Systems, Software Developer | *Python, Flask*

September 2023 – July 2025

- Contributed to a 60+ member team developing autonomous drones; placed 2nd at the 2025 AEAC competition
- Migrated 15+ endpoints in a Flask-based HTTP API connecting backend systems and ArduPilot, to a cross-platform architecture using Pymavlink, eliminating reliance on Windows-only Mission Planner client scripts

TECHNICAL SKILLS

Languages: Python, Golang, TypeScript, Dart, C, C++, C#, Java, R, HTML, SQL

Libraries & Frameworks: FastAPI, Node.js, Express.js, Tailwind CSS, React, JUnit, Mocha, Chai, Jest, Playwright

Developer Tools: Git, GitHub, GitHub Actions, GitLab, Linux, Docker, Docker Compose, Kubernetes, HAProxy, NGINX, Grafana, Cloudflare, GCP, Azure, ChromaDB, PostgreSQL, Swagger